Teacher answer key: Primary 3, Semester 1 (DRAFT)

DRAFT: this answer key is for owner review. Do not give it to pupils. Contact: pypteltd@gmail.com (business email for now, to be changed later)

This key lists the answers and the result of the check on every page. You do not need it for routine marking: a pupil who has a working check has the right answers. To spot-check, compare the check strip with the line "Check result" below.

How the topic tags work

Each entry names the syllabus outcomes the puzzle practises, with the outcome numbers of the Primary 3 content in the Mathematics Syllabus, Primary (2021 edition, pages 35 and 36). The references are for your planning only and are not printed on the pupil pages. "Problem solving" marks a page whose questions come from the kinds found in school tests rather than from one numbered outcome.

Pages in this booklet (Terms 1 and 2)

Strand	Pages
Number and Algebra - Whole Numbers	S1-01, S1-02, S1-03, S1-04, S1-05
Number and Algebra - Addition and Subtraction	S1-06, S1-07, S1-08
Number and Algebra - Multiplication and Division	S1-09, S1-10, S1-11, S1-12, S1-13, S1-14
Number and Algebra - Fractions	S1-15, S1-16, S1-17, S1-18, S1-19
Number and Algebra - Money	S1-20

The page code in the footer, for example S1-07 (P3-07), gives the page in this booklet and, in brackets, its number in the full set of 40.

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S1-01 (P3-01) Place Value 1 star

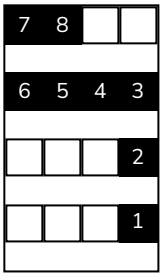
Topic: Place value: thousands, hundreds, tens and ones, numbers to 10 000

Syllabus outcomes: Whole Numbers 1.1, Whole Numbers 1.2. Notes on the tag: 1.1, 1.2 (counting in hundreds; number notation, representations and place values)
Puzzle type: Code-breaker (place value). How the pupil checks: Code word (6 letters).
Answers: 1) 4 thousands 1 hundreds 3 tens 2 ones = 4132; 2) 2 thousands 0 hundreds 2 tens 3 ones = 2023; 3) 14; 4) 7406; 5) 60; 6) 7090
Check result: Code word: QUIVER
Teacher note: Two place-value charts (discs marked 1000, 100, 10 and 1; one chart has a column with no discs), the value of a digit, a regrouped number (ten or more hundreds), a number with a zero hundreds place and the number of hundreds in a number. The wrong numbers a pupil gets by writing the digits in a different order, or the digit instead of its value, have decoy letters in the table or no letter. A box left blank gives no letter and the check cannot see it.

S1-02 (P3-02) Words and Numbers 1 star

Topic: Reading and writing numbers in words and in numerals

Syllabus outcomes: Whole Numbers 1.3. Notes on the tag: 1.3 (reading and writing numbers in numerals and in words)
Puzzle type: Maze / path. How the pupil checks: Route: each step goes to a box whose top words say the numeral at the foot of the box you leave; the line must reach FINISH through the right number of boxes.
Answers: Route (row, column from top-left): (4,4) seven thousand five hundred and seventy-five / 8635 -> (3,4) eight thousand six hundred and thirty-five / 2993 -> (2,4) two thousand nine hundred and ninety-three / 2227 -> (2,3) two thousand two hundred and twenty-seven / 7488 -> (2,2) seven thousand four hundred and eighty-eight / 6445 -> (2,1) six thousand four hundred and forty-five / 9739 -> (1,1) nine thousand seven hundred and thirty-nine / 4390 -> (1,2) four thousand three hundred and ninety / 2297



Check result: Route reaches FINISH through 8 boxes; every top word names the number at the foot of the box before it.
Teacher note: A 4 by 4 grid. Each box has a number in words on top and a numeral at the foot. From a box the pupil moves to a touching box whose top words say the numeral at the foot of the box being left. All numbers are 4-digit. Wrong turns are boxes that look right but are not: digits swapped, teen and ty mixed up, a zero in the wrong place, a hundred or a thousand more or less. Sometimes two touching boxes have the same words on top, so the pupil may meet two choices; only one leads on to FINISH. 2 wrong turns can be entered from the route and lead nowhere, so the box count and the words matter. Pupils read numbers in words and match them with numerals; writing number words by hand is not checked by this page.

S1-03 (P3-03) Digit Cards 2 stars

Topic: Numbers to 10 000: make, compare and order numbers from digit cards

Syllabus outcomes: Whole Numbers 1.2, Whole Numbers 1.4. Notes on the tag: 1.2, 1.4 (place values; comparing and ordering numbers)
Puzzle type: Code-breaker (digit cards). How the pupil checks: Code word (6 letters).
Answers: 1) greatest odd: 8203; 2) greatest: 9861; 3) greatest even: 9736; 4) smallest even: 1348; 5) smallest: 1089; 6) smallest odd: 1369
Check result: Code word: CANOPY
Teacher note: Six sets of four digit cards: greatest, smallest, greatest even, smallest even, greatest odd and smallest odd. A zero card cannot be the thousands digit. The decoy letters stand for the other arrangements of the same cards, so a pupil who makes a different number from the same cards still gets a letter, but not the right one. A box left blank gives no letter and the check cannot see it.

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S1-04 (P3-04) Compare and Order 2 stars

Topic: Numbers to 10 000: comparing and ordering

Syllabus outcomes: Whole Numbers 1.4. Notes on the tag: 1.4 (comparing and ordering numbers)
Puzzle type: Ordering (lettered cards). How the pupil checks: Letters of the cards in the order asked spell a word (6 letters).
Answers: Set A (smallest first): C = 2473, O = 7234, U = 7423; Set B (greatest first): S = 8054, I = 7154, N = 976
Check result: Word: COUSIN
Teacher note: Set A has three 4-digit numbers made from the same four digits, so the pupil must compare place by place from the thousands. Set B has two 4-digit numbers that differ in one digit and one 3-digit number, which is the smallest although its digits look large. The letters, read in the order asked, spell a word with exactly one arrangement that is a word, so no wrong order of the six cards spells a word (every wrong order was tested against a very large word list). A box left blank gives no letter and the check cannot see it.

S1-05 (P3-05) Number Patterns 2 stars

Topic: Number patterns: counting in hundreds and thousands, two-step patterns

Syllabus outcomes: Whole Numbers 1.1, Whole Numbers 1.5. Notes on the tag: 1.1, 1.5 (counting in hundreds and thousands; patterns in number sequences)
Puzzle type: Lettered choice (patterns). How the pupil checks: Word from the letters of the right choices (6 letters).
Answers: 1) 4000; 2) 3290; 3) 4090; 4) 5200; 5) 8500; 6) 4900
Check result: Word: POLICE
Teacher note: Six number patterns with one missing number: counting on and back in hundreds and thousands, and two patterns whose two steps take turns. Each row offers three numbers; the wrong ones are the usual slips (the wrong place changed, or the wrong step). The letters of the right numbers, read down the rows, spell the word. Every other way of choosing was tested against a very large word list: none spells a word. A box left blank gives no letter and the check cannot see it.

S1-06 (P3-06) Big Sums 1 star

Topic: Add and subtract in columns, 4-digit numbers

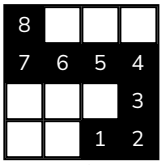
Syllabus outcomes: Addition and Subtraction 2.1. Notes on the tag: 2.1 (addition and subtraction algorithms, up to 4 digits)
Puzzle type: Code-breaker (written methods). How the pupil checks: Code word (6 letters).
Answers: 1) 3651 + 4226 = 7877; 2) 5847 - 3794 = 2053; 3) 5605 - 3816 = 1789; 4) 6269 - 2000 = 4269; 5) 5254 + 1820 = 7074; 6) 5097 + 1667 = 6764
Check result: Code word: TEACUP
Teacher note: Six vertical sums within 10 000: three additions (no renaming, ones and tens renamed, hundreds renamed into the thousands) and three subtractions (no renaming, tens and hundreds renamed, and one with a zero in the top number). The decoy letters stand for the answers a pupil gets by not renaming (adding or subtracting digit by digit). A box left blank gives no letter and the check cannot see it.

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S1-07 (P3-07) Mental Pairs 1 star

Topic: Mental sums: adding and subtracting two 2-digit numbers

Syllabus outcomes: Addition and Subtraction 2.2. Notes on the tag: 2.2 (mental calculation: addition and subtraction of two 2-digit numbers)
Puzzle type: Maze / path. How the pupil checks: Route: the line must reach FINISH through the right number of boxes, every box making 48.
Answers: Route (row, column from top-left): (4,3) 95 – 47 -> (4,4) 85 – 37 -> (3,4) 64 – 16 -> (2,4) 77 – 29 -> (2,3) 97 – 49 -> (2,2) 81 – 33 -> (2,1) 84 – 36 -> (1,1) 90 – 42
Check result: Route reaches FINISH through 8 boxes; every box on it makes 48.



Teacher note: A 4 by 4 grid of sixteen mental sums, each the sum or difference of two 2-digit numbers (every sum and every difference needs renaming, so the pupil must think about tens and ones). The boxes off the route make answers near 48 (37, 39, 46, 59); a slip in the tens or the ones sends the pupil there. 2 extra boxes that do make 48 are dead ends. Counting the boxes shows a line that takes a shortcut.

S1-08 (P3-08) Missing Digits 2 stars

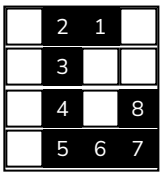
Topic: Add and subtract in columns: find the missing digit

Syllabus outcomes: Addition and Subtraction 2.1. Notes on the tag: 2.1 (addition and subtraction algorithms, up to 4 digits)
Puzzle type: Code-breaker (written methods). How the pupil checks: Code word (6 letters).
Answers: 1) 4997 - 3746 = 1251: digit 4; 2) 8570 - 4311 = 4259: digit 2; 3) 5283 + 4012 = 9295: digit 5; 4) 3651 + 4850 = 8501: digit 8; 5) 7185 - 2975 = 4210: digit 1; 6) 2663 + 7035 = 9698: digit 6
Check result: Code word: FATHOM
Teacher note: Six column sums with one digit missing (three additions, three subtractions, the missing digit in the top number, the bottom number or the answer). Each has exactly one digit that makes it true. The pupil writes the missing digit, then looks it up. A digit that is wrong either has a decoy letter or no letter. A box left blank gives no letter and the check cannot see it.

S1-09 (P3-09) Tables Maze 1 star

Topic: Multiplication and division facts: tables of 6, 7, 8 and 9

Syllabus outcomes: Multiplication and Division 3.1, Multiplication and Division 3.2, Multiplication and Division 3.5. Notes on the tag: 3.1, 3.2 (tables of 6, 7, 8 and 9; multiplying and dividing within the tables)
Puzzle type: Maze / path. How the pupil checks: Route: the line must reach FINISH through the right number of boxes, every missing number being 6.
Answers: Route (row, column from top-left): (1,3) 5 × ? = 30 -> (1,2) 7 × ? = 42 -> (2,2) 24 ÷ ? = 4 -> (3,2) ? × 8 = 48 -> (4,2) 6 × ? = 36 -> (4,3) 42 ÷ 7 = ? -> (4,4) ? × 4 = 24 -> (3,4) 48 ÷ ? = 8
Check result: Route reaches FINISH through 8 boxes; the missing number in every box on it is 6.



Teacher note: A 4 by 4 grid of sixteen multiplication and division facts with one missing number, from the tables of 6 to 9 and the tables they link to (2 to 5). The route follows the boxes whose missing number is 6. The other boxes have missing numbers near 6. 2 extra boxes that do make 6 are dead ends. Counting the boxes shows a line that takes a shortcut.

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S1-10 (P3-10) True or False Tables 1 star

Topic: Multiplication and division facts: tables of 6, 7, 8 and 9

Syllabus outcomes: Multiplication and Division 3.1, Multiplication and Division 3.2, Multiplication and Division 3.5. Notes on the tag: 3.1, 3.2, 3.5 (tables of 6, 7, 8 and 9; multiplying and dividing within the tables; mental calculation)
Puzzle type: True or false table. How the pupil checks: Word from the letters of the right choices (6 letters).
Answers: 1) False (9 groups of 5 is 55.); 2) False (7 × 5 = 42); 3) True (If 7 × 9 = 63, then 63 ÷ 9 = 7.); 4) True (56 ÷ 8 = 7); 5) False (37 is in the 7 times table.); 6) False (6 × 4 = 34)
Check result: Word: PUBLIC
Teacher note: Six statements from the tables of 6, 7, 8 and 9, including one about a times table and one that links a multiplication to a division. Each row has a True choice and a False choice, each with a letter. The letters of the right choices, read down the rows, spell the word. Every other way of choosing was tested against a very large word list: none spells a word. A box left blank gives no letter and the check cannot see it.

S1-11 (P3-11) Remainders 1 star

Topic: Division with remainder

Syllabus outcomes: Multiplication and Division 3.3. Notes on the tag: 3.3 (division with remainder)
Puzzle type: Code-breaker (division with remainder). How the pupil checks: Code word (6 letters).
Answers: 1) 37 ÷ 6: quotient 6, remainder 1; answer 1; 2) 63 ÷ 9: quotient 7, remainder 0; answer 0; 3) 67 ÷ 5: quotient 13, remainder 2; answer 13; 4) 67 ÷ 5: quotient 13, remainder 2; answer 2; 5) 43 ÷ 4: quotient 10, remainder 3; answer 11; 6) 59 ÷ 8: quotient 7, remainder 3; answer 7
Check result: Code word: THRIVE
Teacher note: Six division items, each asking for a different part of the same kind of answer: a remainder, a number of full bags, the number left over when sharing, the least number of boxes (the quotient plus one when there is a remainder), a quotient, and one remainder that is zero. The decoy letters stand for the quotient when the remainder was asked for, the remainder when the quotient was asked for, and the least number of boxes without the extra box. A box left blank gives no letter and the check cannot see it.

S1-12 (P3-12) Fact Families 2 stars

Topic: Multiplication and division: finding the missing number, the loop that closes

Syllabus outcomes: Multiplication and Division 3.1, Multiplication and Division 3.2. Notes on the tag: 3.1, 3.2 (tables of 6, 7, 8 and 9; multiplying and dividing within the tables)
Puzzle type: Chain loop (multiplication tiles). How the pupil checks: Loop: the number in an empty circle must match the circle it is joined to by =, all the way round and back to tile 1.
Answers: Empty circles: tile 1 left = 3, tile 2 left = 7, tile 3 left = 8, tile 4 right = 4, tile 5 left = 4, tile 6 left = 6. Tiles (top = left × right): 21 = 3 × 7, 56 = 7 × 8, 64 = 8 × 8, 32 = 8 × 4, 24 = 4 × 6, 18 = 6 × 3
Check result: Every touching pair matches, all the way round.
Teacher note: Six tiles, each a product at the top and its two numbers below. One number below every tile is empty; the pupil finds it by dividing the top by the other number and may check by multiplying back. The right number of a tile is the same as the left number of the next tile, and the right number of the last tile is the first tile's left number. So every number the pupil writes is tested against a matching circle: a wrong number breaks the loop. At most one link has an empty circle on both sides. In every tile at least one number is from the tables of 6 to 9; all products are within 81.

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S1-13 (P3-13) Multiply and Divide 2 stars

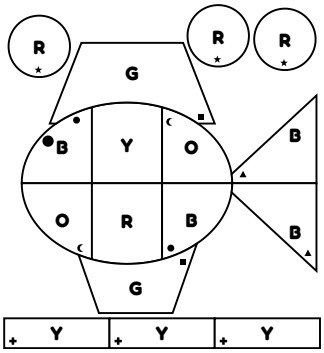
Topic: Multiplication and division of 3-digit numbers by 1-digit numbers

Syllabus outcomes: Multiplication and Division 3.4. Notes on the tag: 3.4 (multiplication and division algorithms, up to 3 digits by 1 digit)
Puzzle type: Code-breaker (written methods). How the pupil checks: Code word (6 letters).
Answers: 1) 568 ÷ 8 = 71; 2) 318 ÷ 3 = 106; 3) 219 × 9 = 1971; 4) 435 × 7 = 3045; 5) 243 × 8 = 1944; 6) 710 ÷ 5 = 142
Check result: Code word: FIGURE
Teacher note: Three multiplications of a 3-digit number by a 1-digit number (renaming in the ones or tens) and three exact divisions of a 3-digit number by a 1-digit number, one of them with a zero in the quotient. The decoy letters stand for the answers a pupil gets by not renaming in a multiplication, or by leaving out the zero of a quotient. A box left blank gives no letter and the check cannot see it.

S1-14 (P3-14) Colour the Fish 2 stars

Topic: Multiplication and division: tables of 6, 7, 8 and 9

Syllabus outcomes: Multiplication and Division 3.1, Multiplication and Division 3.2, Multiplication and Division 3.5. Notes on the tag: 3.1, 3.2, 3.5 (tables of 6, 7, 8 and 9; multiplying and dividing within the tables; mental calculation)
Puzzle type: Colour by code (pupils colour with their own crayons). How the pupil checks: Colour key and tally: every answer must be on the key, the number of parts of each colour must match the box, and parts with the same small mark match.
Answers: Colour key: 6 = yellow, 7 = red, 9 = blue, 24 = green, 36 = orange
Check result: Tally: yellow 4, red 4, blue 4, green 2, orange 2; parts with the same small mark match.



Teacher note: Every part asks a multiplication or division fact from the tables of 6, 7, 8 and 9: a product, a missing number, a division or a missing divisor. Pupils colour the black-and-white outline with their own crayons (or, for a pencil-only copy, write Y, R, B, G or O in each part: the tally and the matching parts can be checked the same way). Check strip: count the parts of each colour (the tally is printed); parts with the same small mark match.

S1-15 (P3-15) Equal Fractions 1 star

Topic: Fractions: equivalent fractions

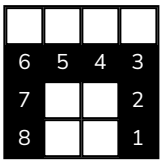
Syllabus outcomes: Fractions 1.1. Notes on the tag: 1.1 (equivalent fractions)
Puzzle type: Lettered choice (equivalent fractions). How the pupil checks: Word from the letters of the right choices (6 letters).
Answers: 1) 4/8; 2) 2/10; 3) 3/9; 4) 6/10; 5) 2/12; 6) 8/10
Check result: Word: SHRIMP
Teacher note: Six figures (bars and circles) cut into equal parts with some parts shaded. Each row offers three fractions and only one is equal to the shaded part (the shaded part written with more, smaller parts). The wrong ones are close in look: the same top with one more part, one more shaded part, or the top and bottom each increased by the same number instead of multiplied. The letters of the right fractions, read down the rows, spell the word. Every other way of choosing was tested against a very large word list: none spells a word. A box left blank gives no letter and the check cannot see it.

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S1-16 (P3-16) Simplest Form 1 star

Topic: Fractions: simplest form

Syllabus outcomes: Fractions 1.2. Notes on the tag: 1.2 (simplest form)
Puzzle type: Maze / path. How the pupil checks: Route: the line must reach FINISH through the right number of boxes, every fraction being in simplest form.
Answers: Route (row, column from top-left): (4,4) 6/7 -> (3,4) 7/8 -> (2,4) 4/9 -> (2,3) 1/10 -> (2,2) 3/10 -> (2,1) 4/5 -> (3,1) 6/11 -> (4,1) 5/11
Check result: Route reaches FINISH through 8 boxes; every fraction on it is in simplest form.



Teacher note: A 4 by 4 grid of sixteen fractions. A fraction is in simplest form when no whole number except 1 divides both the top and the bottom number (so 4/6 is not, 3/5 is). The pupil steps only on those. The boxes off the route are fractions that can still be made simpler (such as 6/8 or 4/12). 2 extra boxes in simplest form are dead ends. Counting the boxes shows a line that takes a shortcut. Denominators are at most 12.

S1-17 (P3-17) Fraction Gaps 2 stars

Topic: Fractions: finding an equivalent fraction when the top or bottom number is given

Syllabus outcomes: Fractions 1.1, Fractions 1.4. Notes on the tag: 1.1, 1.4 (equivalent fractions; finding an equivalent fraction given the denominator or numerator)
Puzzle type: Code-breaker (equivalent fractions). How the pupil checks: Code word (6 letters).
Answers: 1) 5/6 = 10/12; 2) 4/5 = 8/10; 3) 2/5 = 4/10; 4) 3/6 = 6/12; 5) 2/3 = 4/6; 6) 1/3 = 3/9
Check result: Code word: GOPHER
Teacher note: Six pairs of equal fractions with one number missing: three with the top number missing and three with the bottom number missing. The pupil multiplies the top and the bottom by the same number. The decoy letters stand for adding the same number to the top and the bottom instead of multiplying. All numbers are within 12. A box left blank gives no letter and the check cannot see it.

S1-18 (P3-18) Fraction Order 2 stars

Topic: Fractions: comparing and ordering unlike fractions

Syllabus outcomes: Fractions 1.3, Fractions 1.1. Notes on the tag: 1.3 (comparing and ordering unlike fractions; denominators not exceeding 12)
Puzzle type: Ordering (lettered cards). How the pupil checks: Letters of the cards in the order asked spell a word (6 letters).
Answers: Set A (smallest first): J = 2/3, U = 3/4, N = 11/12; Set B (greatest first): I = 7/12, O = 2/4, R = 1/3
Check result: Word: JUNIOR
Teacher note: Each set has three unlike fractions whose denominators are related (for example 2, 4 and 8, or 3, 6 and 12), so the pupil can change them to equal fractions with the same bottom number and compare the tops. No two fractions in a set are equal. The letters, read in the order asked, spell a word with exactly one arrangement that is a word, so no wrong order of the six cards spells a word (every wrong order was tested against a very large word list). A box left blank gives no letter and the check cannot see it.

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S1-19 (P3-19) Fraction Sums 2 stars

Topic: Fractions: adding and subtracting two related fractions within one whole

Syllabus outcomes: Fractions 2.1. Notes on the tag: 2.1 (adding and subtracting two related fractions within one whole)
Puzzle type: Code-breaker (fractions). How the pupil checks: Code word (6 letters).
Answers: 1) 1/8; 2) 5/10; 3) 8/10; 4) 6/10; 5) 2/6; 6) 7/10
Check result: Code word: SCRIBE
Teacher note: Six items with two related fractions, that is, one bottom number is made from the other by multiplying (for example halves and quarters, thirds and ninths). Two additions, two subtractions, one whole minus a fraction and a story. Every answer is smaller than one whole. The bottom number of the answer is printed (the larger one); the pupil changes the other fraction to equal parts and writes the top number. The decoy letters stand for adding or subtracting the top numbers without changing parts, or for the part left over. A box left blank gives no letter and the check cannot see it.

S1-20 (P3-20) Money Sums 1 star

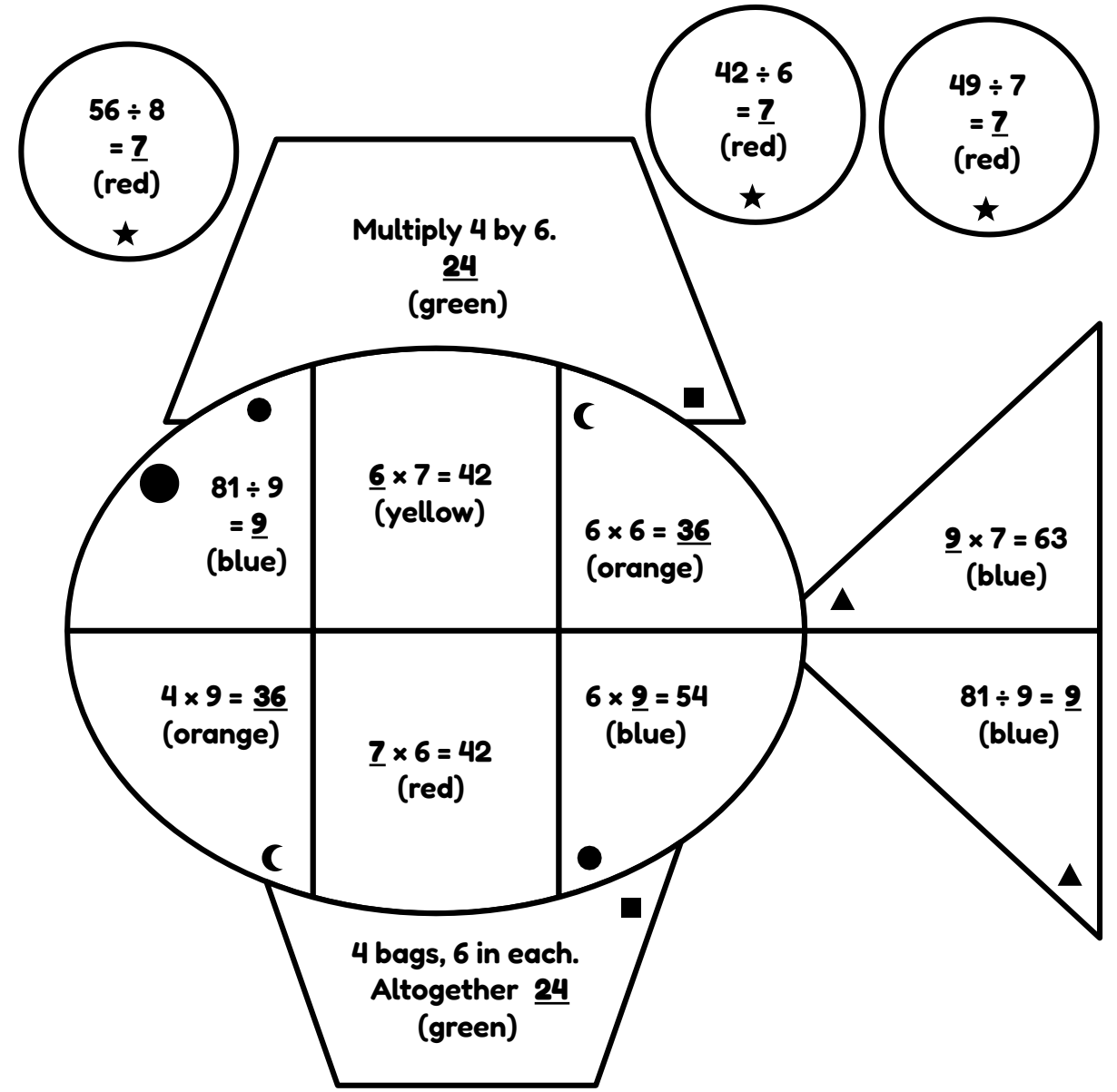
Topic: Money: adding and subtracting in dollars and cents

Syllabus outcomes: Money 1.1. Notes on the tag: 1.1 (adding and subtracting money in decimal notation)
Puzzle type: Code-breaker (money). How the pupil checks: Code word (6 letters).
Answers: 1) $5.70; 2) $8.15; 3) $8.40; 4) $11.80; 5) $1.94; 6) $7.80
Check result: Code word: UPDATE
Teacher note: Six money items in decimal notation: an addition, a subtraction (money left), change from a note, 'how much more', a total of three prices and a count of the notes and coins drawn. Prices are in dollars and cents ($3.45); every answer is written with a dollar sign, a point and two cents digits. The decoy letters stand for adding where the item subtracts, or for a dollar too many or too few. Pictures are not actual size. A box left blank gives no letter and the check cannot see it.

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Answers: S1-14 (P3-14) Colour the Fish

Every part shows the question, the answer underlined and the colour in brackets. Pupils colour their own copy. Colour key: 6 = yellow, 7 = red, 9 = blue, 24 = green, 36 = orange.



48 ÷ 8 = 6 (yellow)	6 × 6 = 36 (yellow)	42 ÷ 7 = 6 (yellow)

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